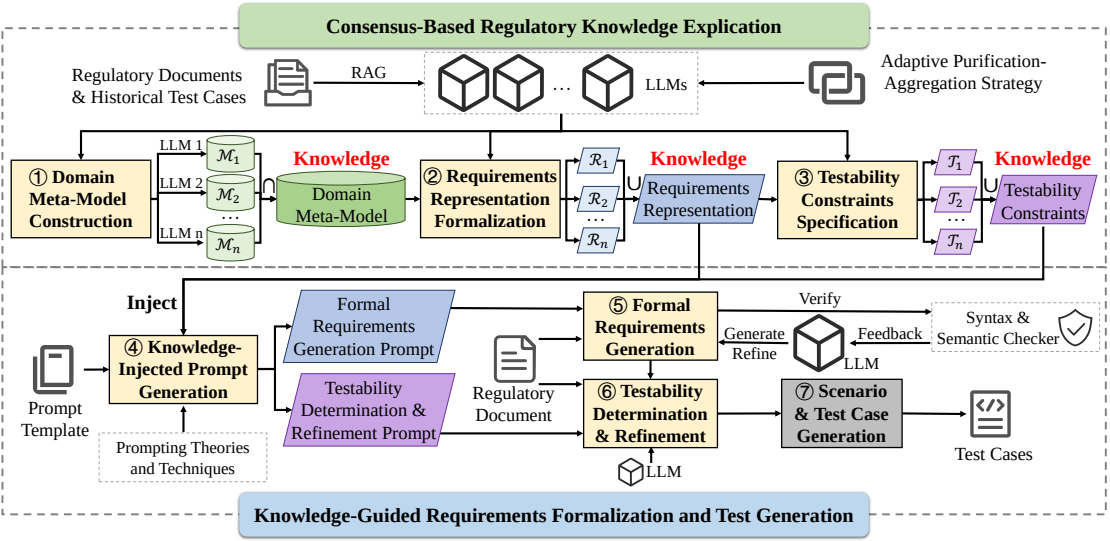

Consensus-Based Regulatory Knowledge Explication
Regulatory Documents & Historical Test Cases
RAG
LLMs
Adaptive Purification-Aggregation Strategy
① Domain Meta-Model Construction
LLM 1
LLM 2
LLM n
Knowledge
Domain Meta-Model
② Requirements Representation Formalization
Knowledge
Requirements Representation
③ Testability Constraints Specification
Knowledge
Testability Constraints
Inject
④ Knowledge-Injected Prompt Generation
Prompt Template
Prompting Theories and Techniques
Formal Requirements Generation Prompt
Testability Determination & Refinement Prompt
Regulatory Document
⑤ Formal Requirements Generation
Verify
Generate
Refine
Feedback
LLM
Syntax & Semantic Checker
⑥ Testability Determination & Refinement
LLM
⑦ Scenario & Test Case Generation
Test Cases
Knowledge-Guided Requirements Formalization and Test Generation